

Baha Eddine Belhaj Mouldi

AI Software Engineer — RAG | LLM | Agentic Systems | FastAPI

Tunis, Tunisia | +216 55 128 982 | bahaeddine.belhajmouldi@gmail.com
linkedin.com/in/bahamouldi | github.com/bahamouldi | bahamouldi.github.io/portfolio

PROFILE

Computer engineering graduate with hands-on experience building production-grade AI systems: RAG pipelines, LLM orchestration, multi-agent architectures and FastAPI services. Comfortable translating business processes into agentic workflows that are testable, auditable and built to last. Proficient in AI-assisted coding (Copilot, Claude) as a daily productivity multiplier. Systems thinker who takes real ownership of what ships. Drawn to applying AI where it removes real operational friction — turning finance and back-office processes into reliable, auditable automation.

CORE SKILLS

RAG & LLM : RAG pipelines (retrieval, generation, evaluation), LangChain, LangGraph, Gemini, Azure AI Foundry, prompt versioning, semantic search, context-aware generation, LLM fine-tuning (LoRA), self-hosted LLMs (Ollama, Llama/Qwen)
Agentic Systems : Multi-agent architectures, MCP (Model Context Protocol) server integrations, tool integration, Agent-to-Agent (A2A) protocols, agentic design patterns, workflow orchestration, testable and auditable agent configs
Backend & APIs : Python, FastAPI, REST APIs, JSON, async services, error logging, observability (monitoring, metrics, quality tracking)
AI & ML : Scikit-learn, Pandas, SHAP, sentiment analysis, text classification, NLP pipelines, AI code generation, prompt engineering
Data & Infrastructure : PostgreSQL, MongoDB, SQL, Azure AI Search, Azure Functions, Docker, Kubernetes, CI/CD (Azure DevOps, Jenkins, ArgoCD, GitHub Actions)
Tools & Process : Git, code review (give and receive), technical documentation, demos & knowledge-sharing, hackathons

PROFESSIONAL EXPERIENCE

AI Software Engineer — Intern (PFE) 01/2026 – 05/2026
Beehive Entreprises, Tunisia

- Designed and shipped production AI systems: RAG chatbot over 900K records, multi-agent architectures and FastAPI services on Kubernetes.
- Built RAG pipelines with LangChain, Gemini and Azure AI Foundry: retrieval, context injection, generation and evaluation — testable agentic workflows.
- Implemented observability: Prometheus/Grafana dashboards, ELK log tracking, quality metrics and real-time alerting for agent performance.
- Versioned prompts and agent configs with code-level rigor. Participated in code reviews on both sides.
- Contributed to AI enablement through demos, workshops and knowledge-sharing sessions.

AI & Backend Developer, Freelance (Part-Time) 01/2025 – 12/2025
Beehive Entreprises, Tunisia

- Built AI-powered backend services: FastAPI, LLM integrations, RAG pipelines, multi-agent orchestration for client applications.
- Code review and vulnerability analysis on 8 web and API applications. Cross-functional collaboration with business teams.

Cybersecurity Intern — SAP Threat Detection 07/2025 – 08/2025
Streamlink, Tunisia

- Automated detection pipeline: Python, ELK correlation, real-time alerting with N8N orchestration. Testable, auditable detection workflows.

Technical Project Lead — Vmate 06/2024
ENIT, Tunisia

- Led team of 4 on full-stack platform — project ranked 1st. Sprint planning, code review, cross-functional collaboration.

KEY PROJECTS

RAG Chatbot — Revenue Assurance Dashboard (Tunisie Telecom) 2026
Python, LangChain, Gemini, Azure AI Foundry, Streamlit, PostgreSQL

- Production RAG pipeline over 900K records across 4 domains: retrieval, context injection, generation with Gemini. Business users query revenue data in natural language. Observability dashboards tracking latency, retrieval quality and generation accuracy.

Multi-Agent Vulnerability Detection System 2025
Python, Multi-Agent, A2A, Azure AI Foundry

- Distributed agent architecture: specialized agents for collection, correlation and alerting. A2A protocols. Testable and auditable workflow design.

Multi-Agent Customer Feedback Analysis 2026
Python, NLP, Sentiment Analysis, ML

- Agentic pipeline: sentiment analysis, topic extraction, actionable insights via multi-agent orchestration. Prompt versioning and config management.

RAG Hotel Recommendation System 2025
Python, LangChain, LLM, Azure AI Foundry

- RAG system: semantic search over reviews, context-aware LLM recommendations, multi-agent orchestration for personalized suggestions.

BeeWAF — Intelligent Web Application Firewall (PFE, graded 86/100) 2026
FastAPI, Python, ML, Docker, Kubernetes, ELK, Prometheus, Grafana

- Enterprise WAF deployed in production: 27 modules, 107 signatures, 3 ML models, CI/CD (Jenkins + ArgoCD), monitoring dashboards, real-time alerting. Score 98.2/100, 0% false positives.

EDUCATION

Engineering Degree in Computer Science 09/2023 – 06/2026
National Engineering School of Tunis (ENIT), Tunisia

- Graduated 06/25/2026. Ranked 65th out of 600+ in the national engineering school entrance exam.

Preparatory Classes (Physics & Technology) 09/2021 – 06/2023
Preparatory Institute for Engineering Studies, El Manar, Tunisia

Technical Baccalaureate — High Distinction 06/2021
Lycee Abdelaziz Khouja, Kelibia, Tunisia

CERTIFICATIONS

Fundamentals of Deep Learning — NVIDIA (2025) ; Exploring Adversarial ML — NVIDIA (2025) ; Machine Learning — NVIDIA (2024) ; Networking — Cisco (2025) ; Git & GitHub — 365 Data Science (2024).

LANGUAGES

French (Fluent) | English (B2) | Arabic (Native) | Chinese (Basic)